(iii) Smoothed Particle Hydrodynamics (SPH): SPH is a mesh-free, particle-based method that is well-suited for simulating free-surface flows and multiphase interactions [see, e.g., Wang *et al.* (2016); Díaz-Damacillo *et al.* (2023); Colagrossi & Landrini (2003); Violeau & Rogers (2016)]. SPH is well-suited for studies of certain types of free-surface flows, but it can be computationally intensive because it requires the tracking of individual particles and, in some cases, stabilization techniques. SPH typically uses additional formulations to capture, accurately, interfacial dynamics, such as surface tension [see Jeske *et al.* (2023)].

(iv) Level Set Method: The Level-Set Method represents interfaces as sharp boundaries using a level-set function. This approach is effective for tracking evolving interfaces, particularly in cases where the topology changes, such as in droplet merging and splitting [see, e.g., Sethian & Smereka (2003); Valle *et al.* (2020); Yuan *et al.* (2018)]. A detailed comparative study via numerical simulations of both Level-Set and phase-field methods is given in Zhu *et al.* (2021), where the authors write, "...In the rising process of the bubble, the tracking efficiency of the phase field method is higher than that of the level set method. The phase field method is easier to capture the subtle changes of the bubble, and has higher calculation efficiency and accuracy, and has better simulated calculation results."

The remaining part of this paper is organised as follows. In Section 2 we begin with a discussion of the statistical mechanics of systems in which two coexisting phases, distinguished from each other by a scalar order parameter ϕ, are separated by an interface. Our discussion leads naturally to the time-dependent Ginzburg-Landau (TDGL) partial differential equation (PDE), when ϕ is *not conserved*, and the Cahn-Hilliard (CH) PDE, if ϕ is *conserved*. In Section 3 we define the models that we use when the coexisting phases are fluids; in the simple case of two immiscible fluids we have the Cahn-Hilliard-Navier-Stokes (CHNS) equations; we give its generalisations for (a) coexisting phases with different viscosities, (b) CHNS with gravity, (c) three-component fluids (CHNS3), and (d) CHNS for active fluids. Section 4 gives an overview of the numerical schemes that are used to study these models; we include details of pseudospectral direct numerical simulations (DNSs) and the volume-penalisation method that we use in our work; furthermore, we contrast the CHNS diffuse-interface approach with schemes that track the spatiotemporal evolution of sharp interfaces. Section 5 discusses mathematical issues of the regularity of solutions of the CHNS PDEs. Section 6 contains a survey of the types of results that have been obtained by numerical studies of CHNS-type PDEs. In Section 7 we present results for a variety of challenging problems for which we have to go beyond the binary-fluid CHNS. The concluding Section 8 summarises the CHNS framework for multiphase flows and discusses and new challenges in this area.

2. Overview: Statistical Mechanics of Interfaces

This Section contains a short outline of the statistical mechanics of systems with interfaces. We include material that is required for the development of Cahn-Hilliard models and their generalisations. Subsection 2.1 is devoted to the equilibrium statistical mechanics of systems with interfaces; in particular, it uses the Ising and lattice-gas models to explore interfacial statistical mechanics. In Subsection 2.2 we turn to a discussion of time-dependent phenomena, especially in the context of the Kinetic Ising model. This leads naturally to the discussion, in Subsection 2.2.1, of time-dependent Ginzburg-Landau theory, for a non-conserved order parameters, and, in Subsection 2.2.2, the Cahn-Hilliard equation for a conserved order parameter.

System	↑ Phase	↓ Phase	Phase Transition
Ising Ferromagnet	ferromagnetic up	ferromagnetic down	spin flip
Liquid-Gas	high-density (liquid)	low-density (gas)	condensation
Binary (A-B) Mixture	A-rich	B-rich	phase-separation
Active Fluid	high-density	low-density	active phase-separation

TABLE 1. Correspondences between Ising ferromagnetic systems, with lattice-gas or binary-mixture counterparts, and their phases and phase transitions; here, ↑ and ↓ represent the up-spin and down-spin phases of an Ising ferromagnet (see text). Rows $2-4$ mention equilibrium phases and transitions; the last row mentions nonequilibrium active fluids which can exhibit active phase-separation [see Section 7.6].

2.1. *Equilibrium Statistical Mechanics*

Interfaces separate coexisting phases. In *equilibrium*, all thermodynamic properties of interfaces follow from the *intensive interfacial free energy* f_I, which is precisely the interfacial tension $\sigma_{\alpha,\beta}$ between coexisting bulk phases α and β. For pedagogical reasons, we illustrate this by considering the ferromagnetic, spin-half, Ising model, which is used to model magnets that are anisotropic (in spin space). This Ising model, which can be mapped onto a simple lattice-gas model for a fluid that can have two coexisting bulk phases (see below), has the Hamiltonian

$$\mathcal{H} = -J \sum_{\langle i,j \rangle} S_i S_j - h \sum_i S_i \,. \tag{2.1}$$

The spins $S_i = \pm 1$ occupy the $N \sim L^d$ sites i of a $d-dimensional$ hypercubic lattice, with side L, in a region Ω; the exchange interaction J couples spins on nearest-neighbour pairs of sites $\langle i,j \rangle$; a ferromagnetic interaction ($J > 0$) favours spin alignment, i.e., spins of the same sign on nearest-neighbour sites; the external magnetic field h favours $S_i = sgn(h)$ and it distinguishes between the two phases: up-spin ↑ (with $S_i = +1\ , \forall i$, at zero temperature $T = 0$) and down-spin ↓ (with $S_i = -1\ , \forall i$, at $T = 0$). If we are interested only in bulk statistical physics, it is convenient to use periodic boundary conditions in all d directions.

The transformation $n_i = (1 + S_i)/2$, yields the lattice gas variables $n_i = 0 \,\text{or}\, 1$ and defines the lattice-gas model, whose low- and high-density phases are the analogues of the ↓ and ↑ phases in the ferromagnetic Ising model and in which the chemical potential is the counterpart of h in the Ising model [see, e.g., Griffiths (1972); Goldenfeld (2018)].† Just as a positive (negative) external field h favours the up-spin ↑ (down-spin ↓) phase, a large and positive (negative) values of the chemical potential favour the high-density (low-density) phase in the lattice-gas model [see Table 1].

The intensive bulk free energy f_B of the Ising model, which is a function of the

† A similar transformation relates the Ising-model Hamiltonian to a lattice model for a binary alloy.

temperature T, h, and J, is defined as follows:

$$\begin{aligned} f_B(T,h,J) &= \lim_{N\to\infty} \frac{1}{N}\Big[F(T,h,J,\Omega)\Big]\,; \\ F(T,h,J,\Omega)] &= -k_B T \ln[\mathcal{Z}]\,; \\ \mathcal{Z} &= \sum_{\{S_i=\pm 1\}} \exp[-\mathcal{H}/(k_B T)]\,; \end{aligned} \tag{2.2}$$

here, k_B, F, and $\mathcal{Z}$ are, respectively, the Boltzmann constant, the total free energy and the partition function, and the sum is over all the spin states. Note that F depends on Ω and, therefore, on boundary conditions; by contrast, f_B does not depend on Ω, the boundary conditions, and boundary couplings and fields because we have taken the *thermodynamic limit* $N \to \infty$. If the thermodynamic limit exists (as it does for the Ising model we consider), then: (a) f_B is a *convex-up* function of its arguments, which guarantees continuity of f_B with respect to T, h, and J; (b) $\partial f_B/\partial T\,, \partial f_B/\partial h$, and $\partial f/\partial J$ exist almost everywhere (f_B has slope discontinuities at first-order phase boundaries); and (c) $\partial f_B/\partial T$, $\partial f_B/\partial h$, and $\partial f/\partial J$ are, respectively, monotone, non-increasing functions of T, h, and J. Bulk thermodynamic functions follow from derivatives of f_B; e.g., the magnetization per site

$$M = \sum_i \langle S_i\rangle/N = -\partial f_B/\partial h\,, \tag{2.3}$$

where the angular brackets denote the thermal average. When the second derivatives of f_B exist, they lead to stability conditions such as the positivity of the specific heat at constant h, to wit, $C_h \equiv -k_B T \partial^2 f_B/\partial T^2 \geqslant 0$.

This magnetization $M \in [-1,1]$ is the *order parameter*, which is positive (negative) in the up-spin (down-spin) phase; its lattice-gas counterpart is the density $\rho = \sum_i \langle n_i\rangle/N \in [0,1]$. For dimension $d > 1$, the Ising-model phase diagram in the $h/J - k_B T/J$ plane (henceforth, we set $k_B = 1$ and $J = 1$) shows a first-order phase boundary, $h = 0$ for $T < T_c$, where T_c is the critical or Curie temperature at which this model exhibits a continuous phase transition from the ferromagnetic to the paramagnetic phase [left panel of Fig. 2]. Along the first-order boundary, the $\uparrow$ and $\downarrow$ phases coexist. In the $T-M$ counterpart [right panel of Fig. 2] of the $h-T$ phase diagram, two-phase coexistence occurs everywhere below the *coexistence curve* in the peach-shaded region: the system undergoes macroscopic phase separation, forming distinct $\uparrow$ and $\downarrow$ phases, rather than a fully mixed state; in equilibrium, these coexisting phases are separated by an interface (see below). If we use the Ising-model-lattice-gas correspondence [Table 1], we see that its binary-mixture counterpart is the coexistence of immiscible fluids, below the critical temperature above which the two fluids mix.

We do not concentrate on the critical properties of the Ising model here; but we do mention critical-point behaviours occasionally. Therefore, we note that, in the vicinity of the critical point at $h = 0$ and $T = T_c$, thermodynamic functions display power-law forms, with universal exponents and scaling functions [see, e.g., Goldenfeld (2018); Kardar (2007); Chaikin *et al.* (1995)]. For example, at $h = 0$ and $(T - T_c)/T_C \ll 1$, $M \sim [(T_c - T)/T_c]^\beta$, for $T \lesssim T_c$; and the specific heat $C_h \sim [(T_c - T)/T_c]^{-\alpha}$ diverges; β and α are universal critical exponents that depend on d and the number of components n of the order parameter (for the Ising model we have a scalar order parameter and $n = 1$). If we were to obtain order-parameter correlation functions and, therefrom, the correlation length ξ, then we would find the divergence $\xi \sim [(T_c - T)/T_c]^{-\nu}$. [At such a critical point, only two exponents are independent; the others can be related by scaling laws that are

Oki K & T (1977); Gunton *et al.* (1983) and Fig. 3 in Gunton *et al.* (1983), which has been reproduced from Oki K & T (1977)]. At the level of Landau-Ginzburg theory, the loci of points along which $g''(\phi) = 0$ is the *spinodal curve* that provides a clear boundary between the regions with nucleation-dominated ($g''(\phi) > 0$) growth and spinodal decomposition ($g''(\phi) < 0$) in Fig. 4 (a). If we go beyond mean-field or Landau-Ginzburg theory and consider the early stages of phase separation, the sharp spinodal curve is replaced by a crossover regime across which the phase-separating system moves from nucleation-dominated to spinodal-decomposition-initiated growth [see, e.g., Oki K & T (1977); Gunton *et al.* (1983)].

• Late stages of phase separation: Here, we must distinguish Lifshitz-Slyozov and Lifshitz-Allen-Cahn scaling laws, for conserved and non-conserved ϕ, respectively; the former [latter] leads to domain growth in time t that is characterised by the power-law growth of the length $\mathcal{L}(t) \sim t^{1/3}$ [$\mathcal{L}(t) \sim t^{1/2}$] as discussed, e.g., in Puri (2004); Puri & Wadhawan (2009); Bray (2002); Lifshitz (1962); Lifshitz & Slyozov (1961); Fogedby & Mouritsen (1988); hydrodynamical effects can modify the domain-growth power-law exponent [see, e.g., Puri (2004); Puri & Wadhawan (2009); Bray (2002)].

3. Cahn-Hilliard-Navier-Stokes Models

Henceforth, we will not consider critical phenomena in the Cahn-Hilliard-Navier-Stokes (CHNS) system and its generalisations. We will concentrate on turbulence in these systems, well below the transition temperature T_c, and the effects it has on the suppression of phase separation (also called coarsening arrest) or droplet motion in turbulent binary- or ternary-fluid flows, the coalescence of droplets or lenses, and turbulence and self-propelled droplets in active-CHNS models. Therefore, it will be convenient to work at a fixed temperature below T_c and use a Landau-Ginzburg free-energy functional in which parameters are scaled in such a way that the minima of F_{LG} are at $\phi_b = \pm 1$ in the two-fluid case. For related derivations of the CHNS model and reviews of diffuse-interface models see, e.g., Gurtin *et al.* (1996) and Anderson *et al.* (1998). We define below the CHNS models that we consider. Subsections 3.1 and 7.1 cover, respectively, binary- and ternary-fluid systems. In Subsection 7.1.1 describes the Boussinesq approximation for the ternary-fluid case. We then turn to active models: In Subsection 7.6 we introduce the active Model H and in Subsection 7.7 a generalisation that allows us to study the self propulsion of an active droplet.

3.1. *Binary-fluid CHNS*

(i) For the binary-fluid case [see, e.g., Guo *et al.* (2021); Borcia *et al.* (2022)] we use the following free-energy functional for our CHNS equations:

$$\mathcal{F}[\phi, \nabla\phi] = \int_\Omega d\Omega \left[\mathcal{V}(\phi) + \frac{3}{4}\sigma\epsilon|\nabla\phi|^2 \right], \tag{3.1}$$

where $\mathcal{V}(\phi)$ is the potential, σ is the bare interfacial tension, and ϵ is the width of the interface. The following two forms of $\mathcal{V}(\phi)$ are used in the CHNS literature:

(a) Curie-Weiss-type potential:

$$\mathcal{V}_{CW}(\phi) = \frac{a}{2}\left[(1+\phi)\ln(1+\phi) + (1-\phi)\ln(1-\phi)\right] - \frac{b}{2}\phi^2\,; \tag{3.2}$$

this potential has the virtue that $\phi \in [-1, 1]$. [In some studies $\mathcal{V}_{CW}$ is called the Flory-Huggins logarithmic potential [see, e.g., Giorgini & Temam (2020)], as in the theory of polymer solutions [see, e.g., Rubinstein & Colby (2003)].]

Dimensionless numbers	Formulae
Reynolds number	$\mathrm{Re} = U_0 L_0/\nu$
Peclet number	$\mathrm{Pe} = U_0 L_0/\nu$
Bond number	$\mathrm{Bo} = \mathcal{A}\rho L_0^2 g/\sigma$
Ohnesorge number	$\mathrm{Oh} = \nu\sqrt{\frac{\rho}{\sigma L_0}}$
Capillary number	$\mathrm{Ca} = \rho\nu U_0/\sigma$
Weber number	$\mathrm{We} = \rho U_0^2 L_0/\sigma$
Cahn number	$\mathrm{Cn} = \epsilon/L_0$

TABLE 2. Important dimensionless numbers for the binary-fluid CHNS system

(b) Landau-Ginzburg-type ϕ^4 potential:

$$\mathcal{V}_{LG}(\phi) = \frac{3}{16}\frac{\sigma}{\epsilon}(\phi^2-1)^2\,; \tag{3.3}$$

this parametrization has the virtue that the global minima of $\mathcal{V}_{LG}(\phi)$ are at $\phi_b = \pm 1$. However, now $\phi \in [-\infty, \infty]$. We use $\mathcal{V}_{LG}(\phi)$ in most of our CHNS studies.

(ii) If we allow the shear viscosity η and the density ρ to depend on ϕ, so that the two coexisting phases have different viscosities (η_1 and η_2) and densities (ρ_1 and ρ_2), far away from interfaces, the incompressible CHNS equations can be written as follows:

$$\begin{aligned}
\rho(\phi)(\partial_t \boldsymbol{u} + (\boldsymbol{u}\cdot\nabla)\boldsymbol{u}) &= -\nabla P + \nabla\cdot\left[\eta(\phi)(\nabla\boldsymbol{u} + \nabla\boldsymbol{u}^T)\right] \\
&\quad - \phi\nabla\mu + \rho(\phi)\boldsymbol{g} - \alpha\boldsymbol{u} + \boldsymbol{f}^{ext}\,; \\
\nabla\cdot\boldsymbol{u} &= 0\,; \\
\partial_t\phi + (\boldsymbol{u}\cdot\nabla)\phi &= M\nabla^2\mu\,; \\
\mu &= \frac{\delta\mathcal{F}}{\delta\phi}\,; \\
\eta(\phi) &= \eta_1\left(\frac{1+\phi}{2}\right) + \eta_2\left(\frac{1-\phi}{2}\right)\,; \\
\rho(\phi) &= \rho_1\left(\frac{1+\phi}{2}\right) + \rho_2\left(\frac{1-\phi}{2}\right)\,.
\end{aligned} \tag{3.4}$$

Here, $\boldsymbol{g}$, $\boldsymbol{f}^{ext}$, and α are, respectively, the acceleration because of gravity, an external forcing, and the coefficient of friction (that is required typically in two-dimensional settings, e.g., to account for bottom friction).

(iii) Boussinesq approximation: If the density difference between the two phases is not too large, in the Navier-Stokes part of the above equations we can use the Boussinesq approximation [see, e.g., Lee & Kim (2013); Celani *et al.* (2009); Boffetta *et al.* (2010*a*,*b*); Zanella *et al.* (2020); Khan & Shah (2019); Shah *et al.* (2017); Forbes *et al.* (2022); Huang & Yang (2022)]. Furthermore, this approximation holds only when the vorticity is not

too strong; otherwise, non-Boussinesq effects, such as interfacial instabilities, can emerge even at low Atwood numbers [see, e.g., Ramadugu *et al.* (2022)]. In this Boussinesq approximation the Navier-Stokes equations can be written as follows:

$$\partial_t \boldsymbol{u} + (\boldsymbol{u}\cdot\nabla)\boldsymbol{u} = -\frac{1}{\rho_0}\nabla P + \nu\nabla^2\boldsymbol{u} - \frac{1}{\rho_0}(\phi\nabla\mu) + \frac{[\rho(\phi)-\rho_0]}{\rho_0}\boldsymbol{g} - \alpha\boldsymbol{u}\,,$$
$$\nabla\cdot\boldsymbol{u} = 0\,, \tag{3.5}$$

where the mean density is $\rho_0 = (\rho_1 + \rho_2)/2$. This approximation neglects density differences except in the term with *gravity*. We can write

$$\frac{(\rho(\phi)-\rho_0)}{\rho_0} = -\frac{\Delta\rho}{\rho_0}\phi \equiv -\mathcal{A}\phi\,, \tag{3.6}$$

with the Atwood number $\mathcal{A} \ll 1$.

In Table 2 we give the important dimensionless parameters that govern the states of the binary-fluid CHNS system. The larger the Reynolds number Re, the more the inertia-induced turbulence, which can lead, e.g., to coarsening arrest [Subsection 6.3]. The Bond number Bo plays a crucial role in gravity-driven flows that include the evolution of antibubbles [Subsection 6.4] or a bubble passing through a liquid-liquid interface [Subsection 7.4]; the Ohnesorge number Oh is important in the coalescence of droplets or liquid lenses [Subsection 7.5]; and the Weber (We) and Capillary (Ca) numbers affect interface-induced low-Re turbulence [Subsection 6.5]. The Cahn number governs the thickness of the interface. Other dimensionless numbers must be introduced as we increase the complexity of the multi-phase system; e.g., in three-phase flows, density, surface-tension, and viscosity ratios are relevant [Subsection 7.2], and the activity parameter is important in active CHNS systems [Subsection 7.8 and 7.9].

4. Numerical Methods

Various numerical methods have been employed successfully to simulate multiphase flows using phase-field methods, in general, and the CHNS equations, in particular. To capture the underlying physics, it is crucial to resolve interfaces accurately and to track their evolution. Common numerical discretization techniques include the finite element method (FEM) [see, e.g., Vey & Voigt (2007); Lowengrub *et al.* (2009); Barrett *et al.* (1999, 2001); Elliott & French (1987)], finite difference methods [see, e.g., Teigen *et al.* (2011); Furihata (2001); Kim & Kang (2009)], and the Lattice-Boltzmann method (LBM) [see, e.g., Timm *et al.* (2016); Chen & Doolen (1998); Benzi & Succi (1990); Benzi *et al.* (1992); Shan & Chen (1993, 1994); Scarbolo & Soldati (2013)], each one of which offers distinct advantages that depend on both the complexity of the problem and the computational demands of the numerical simulation. In this article, we adopt direct numerical simulations (DNSs) that are based on the pseudospectral approach [see, e.g., Canuto *et al.* (1988)].

Direct numerical simulations (DNSs) play an important role in obtaining solutions of the CHNS equations. Interfaces are diffuse in CHNS systems, so we do not have to impose boundary conditions on complicated interfaces that evolve in time. A DNS must, of course, accurately resolve all relevant scales of motion, given initial and boundary conditions. Such DNSs can achieve a high level of accuracy by retaining the governing equations in their complete (rather than reduced) forms. However, DNSs can be computationally expensive, particularly at high Reynolds numbers and for coupled

$\nu = 3.5 \times 10^{-3}$, $g = 1$, $\mathcal{A}_1 = 0.132$, $\mathcal{A}_2 = 0.132$, $\rho_1 = 1.132, \rho_2 = 0.868, \rho_3 = 1$ [with ρ_3 as the reference density], and $\sigma_{12} = \sigma_{13} = 0.5$, and $\sigma_{23} = 0.01$. Figures 15 (a) and (b) show, respectively, isosurface plots of the c_1 and c_2 fields and isocontour plots of the z component of the vorticity field, which illustrate how a c_1 droplet of the first fluid (in red), with concentration field c_1, passes through the $c_2 - c_3$ interface (light brown) between the second and third fluids. As it passes through the interface, this droplet entrains some of the heavy fluid, which forms a slender neck that collapses eventually to yield droplets of the the heavy fluid that fall back onto the interface [see, e.g., Singh & Bart (2015) and Emery *et al.* (2018)]. In Fig. 15(c) we show that we can track the passage of this bubble through the interface by monitoring the temporal evolution of the droplet's kinetic energy [$e_d(t) \equiv \langle |\boldsymbol{u}(\boldsymbol{x})|^2 \rangle_{\boldsymbol{x} \in (c_1 \geqslant 0.5)}$ (blue line)] and the energy of the $c_2 - c_3$ interface [$e_I(t) = \langle |\boldsymbol{u}(\boldsymbol{x})|^2 \rangle_{\boldsymbol{x} \in (0.1 \leqslant c_2 \leqslant 0.9)}$ (red line)]; both these quantities display maxima when the bubble passes through the interface.

7.5. *The coalescence of liquid lenses and droplets*

The coalescence of liquid droplets and lenses is a problem of fundamental importance in fluid mechanics and statistical mechanics [see, e.g., Paulsen *et al.* (2014); Pal (2016); Heinen *et al.* (2022); Padhan & Pandit (2023*b*); Scheel *et al.* (2023)]. Our DNS for the CHNS3 [Padhan & Pandit (2023*b*)] model can be used to examine the development of liquid-lens mergers in phase-separated ternary-fluid systems as we summarise below. The coexistence of three immiscible fluids leads to three distinct interfaces with three interfacial tensions: σ_{ij} is the surface-tension coefficient for the ij interface, where the integers i and j ($= 1, 2,$ or 3) label the coexisting phases. We prepare neutrally buoyant, symmetrical or asymmetrical, lenses in 2D by starting our DNS with the following configuration for a single circular droplet of fluid 1, with radius R_0 and centre at (π, π), placed at the interface between fluids 2 and 3 (see Fig. 16(a)):

$$\begin{aligned} c_1(x, y, 0) &= \frac{1}{2}\left[1 - \tanh\left(\frac{\sqrt{(x-\pi)^2 + (y-\pi)^2} - R_0}{2\sqrt{2}\epsilon}\right)\right]; \\ c_2(x, y, 0) &= \frac{1}{2}\left[1 - \tanh\left(\frac{y-\pi}{2\sqrt{2}\epsilon}\right)\right] - c_1(x, y, 0); \\ \omega(x, y, 0) &= 0. \end{aligned} \tag{7.12}$$

The initial and equilibrium configurations in 3D are a sphere and a lenticular biconvex lens, respectively. As time evolves in our (2D) DNS, the initial circular droplet relaxes to its equilibrium-lens shape as shown in Figs. 16(b) and (c) for $(\sigma_{12}, \sigma_{13}, \sigma_{23}) = (1, 1, 1)$ and $(\sigma_{12}, \sigma_{13}, \sigma_{23}) = (1.4, 0.8, 1)$, respectively. Now we verify the Young relations for liquid lenses at equilibrium [see, e.g., McHale *et al.* (2022); Boyer & Lapuerta (2006)]. The theoretical distance d^{th} between the two triple-phase junctions in Figs. 16 (b) and (c) are given by the following Young relations:

$$\begin{aligned} d^{th} &= (l_1 + l_2)^{-\frac{1}{2}}; \\ l_1 &= \frac{2(\pi - \theta_3) - \sin(2(\pi - \theta_3))}{8A \sin(\pi - \theta_3)}; \\ l_2 &= \frac{2(\pi - \theta_1) - \sin(2(\pi - \theta_1))}{8A \sin(\pi - \theta_1)}, \end{aligned} \tag{7.13}$$

The contact angles are related as follows:

$$\frac{\sin\theta_1}{\sigma_{23}} = \frac{\sin\theta_2}{\sigma_{13}} = \frac{\sin\theta_3}{\sigma_{12}}. \tag{7.14}$$

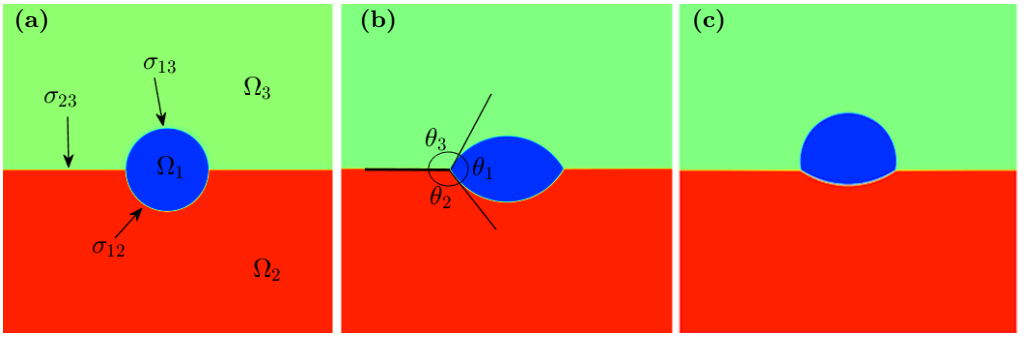

FIGURE 16. Plots showing the three coexisting phases 1 (blue), 2 (red), and 3 (green) in regions Ω_1, Ω_2, and Ω_3, respectively: (a) The intial profile with a circular droplet of radius $R_0/L = 0.15$ of fluid 1 at the interface between fluids 2 and 3. [$L = 2\pi$ is the side length of the simulation domain.] (b) The equilibrium profile of the droplet is a symmetrical lens, because we choose $(\sigma_{12}, \sigma_{13}, \sigma_{23}) = (1, 1, 1)$; the three contact angles are θ_1, θ_2, and θ_3. (c) The equilibrium profile of the droplet is an asymmetrical lens if we choose $(\sigma_{12}, \sigma_{13}, \sigma_{23}) = (1.4, 0.8, 1)$.

Runs	$(\sigma_{12}, \sigma_{13}, \sigma_{23})$	d^{th}	d^{sim}	Relative error (%)
$\mathcal{R}1$	(1, 1, 1)	2.15	2.18	1.4
$\mathcal{R}2$	(1.4, 0.8, 1)	1.3	1.35	3

TABLE 3. The distance between two triple-phase junctions calculated from theory d^{th} [see Eq. (7.14)] and numerical simulations d^{sim} for the symmetrical lens (Fig. 16(b)) and the asymmentrical lens (Fig. 16(c)).

We calculate the distance between the triple-phase junctions d^{sim} from our simulations [see Figs. 16(b) and (c)] and compare them with d^{th} in Table 3. The agreement between these values is good. Furthermore, we can compare the results of our DNS with the prediction of the Laplace law for pressure jumps, at equilibrium; these jumps are defined as folllows:

$$\frac{\sigma_{13}}{R_{13}} = P_1 - P_3 = P_1 - P_2 = \frac{\sigma_{12}}{R_{12}}\,; \quad P_2 - P_3 = 0\,; \tag{7.15}$$

R_{12} and R_{13} are the radii of curvatures of the interfaces between fluids 1 and 2 and between fluids 1 and 3, respectively. To calculate the radii of curvatures, we use the circle Hough transform (CHT) in MATLAB [see Atherton & Kerbyson (1999)]. In Fig. 17, we illustrate the CHT of the images in Figs. 16(b) and (c); red circles are the best-fit circles to the curves. The CHT gives the coordinates of the centres and the radii of the red circles. We calculate the theoretical values of the pressure jumps from these radii of curvatures. To compare our DNS results with these theoretical values, we evaluate the pressure jumps from the following pressure-Poisson equation:

$$\nabla^2 P = \boldsymbol{\nabla} \cdot \left(-\sum_{i=1}^{3} c_i \boldsymbol{\nabla} \mu_i \right) , \tag{7.16}$$

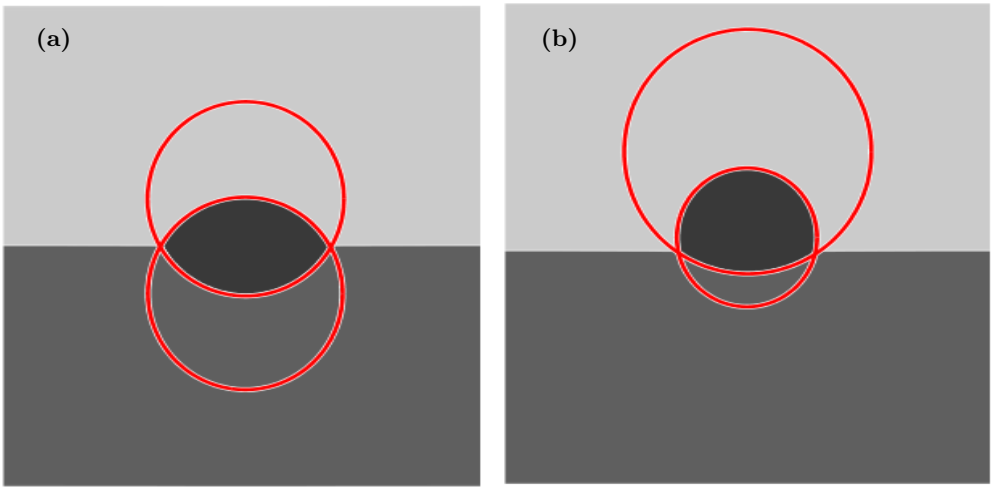

FIGURE 17. Illustrations of the circle Hough transform (see text) that we use to fit circles for the lens interfaces for (a) $(\sigma_{12}, \sigma_{13}, \sigma_{23}) = (1, 1, 1)$ (run $\mathcal{R}1$) and (b) $(\sigma_{12}, \sigma_{13}, \sigma_{23}) = (1.4, 0.8, 1)$ (run $\mathcal{R}2$).

TABLE 4. Illustrative comparisons of theoretical and our DNS results for Laplace-pressure jumps for different lens shapes. $\Delta P \equiv P_1 - P_2 = P_1 - P_3$. There is good agreement between these results. While evaluating ΔP, we calculate the values of P_1, P_2, and P_3 at points that are far from the interface.

Runs	$(\sigma_{12}, \sigma_{23}, \sigma_{13})$	R_0/L	Theory ΔP	Simulation ΔP	Relative error(%)
RN1	(1, 1, 1)	0.2	0.989	0.994	0.5
RN2	(1, 0.8, 1)	0.2	1.148	1.132	1.3
RN3	(0.6, 1, 0.8)	0.2	0.465	0.460	1
RN4	(1.4, 1, 0.6)	0.2	0.889	0.891	0.2
RN-3D	(1, 1, 1)	0.13	1.54	1.56	1.2

which we portray via pseudocolor plots of illustrative equilibrium-pressure profiles for runs RN1 and RN2 in Fig. 18. We present our DNS results in TABLE. 4 for various runs. These results are in good agreement with their theoretical counterparts. We obtain similar results from 3D DNSs of the CHNS3 model: In Fig. 19(a) we give an isosurface plot, with $c_1 = 0.5$, for a 3D lenticular biconvex lens. We then calculate its Gaussian curvature κ by implementing, in MATLAB, the algorithm described in Meyer *et al.* (2003). The isosurface plot of κ is shown in Fig. 19(b); clearly, κ is constant throughout the lens surface, except at the edges. So, in Fig. 19(c), we present the probability distribution function (PDF) of κ to find out the most probable value of κ. We consider the values of κ with the highest probability, namely, $0.59, -0.04, 1.23$; the average value is $\kappa \simeq 0.593$.

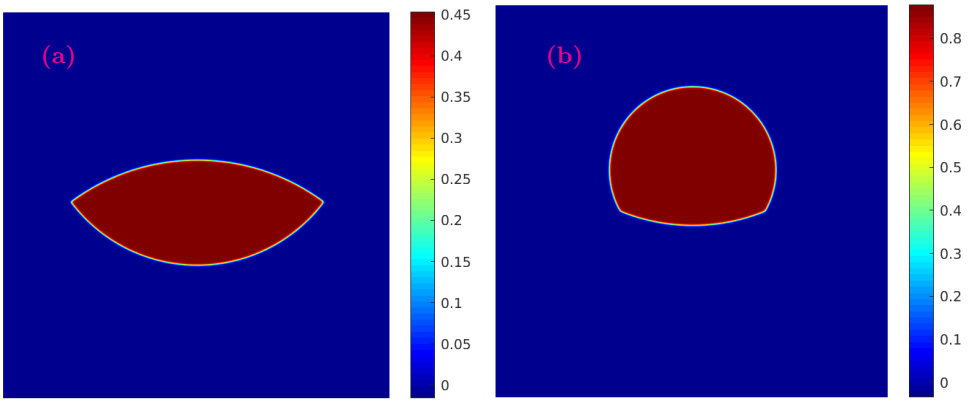

FIGURE 18. Pseudocolor plot of illustrative equilibrium-pressure profiles for (a) $(\sigma_{12}, \sigma_{23}, \sigma_{13}) = (0.6, 1, 0.8)$ (run RN3) and (b) $(\sigma_{12}, \sigma_{23}, \sigma_{13}) = (1.4, 1, 0.6)$ (run RN4).

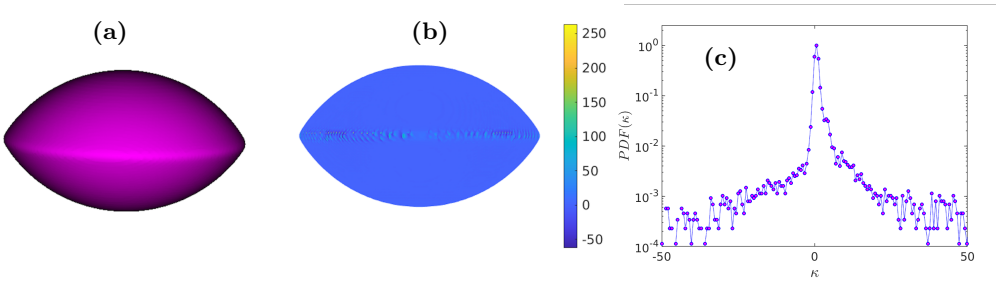

FIGURE 19. (a) The isosurface plot of $c_1 = 0.5$, at equilibrium, illustrating a lens in 3D. (b) The isosurface plot of the corresponding Gaussian curvature κ. (c) The PDF of the Gaussian curvature.

The Gaussian curvature $\kappa = 1/R_G^2$ for a sphere of radius R_G [see, e.g., Nothard *et al.* (1996)]. The symmetric 3D lens in Fig. 19(a) is a combination of two surfaces that are parts of spheres of equal radii. Then we follow the Laplace law in 3D to evaluate the pressure jumps:

$$\frac{2\sigma_{13}}{R_{13}} = P_1 - P_3 = P_1 - P_2 = \frac{2\sigma_{12}}{R_{12}}\,;$$
$$P_2 - P_3 = 0\,; \tag{7.17}$$

here $R_{13} = R_{12} = R_G = 1/\sqrt{\kappa}$. The theoretical values of the pressure jumps are given in Table 4 (see run RN-3D); we solve the pressure-Poisson equation (7.16) numerically; we find good agreement between the theoretical and numerical values of these jumps .

We turn now to an overview of our recent study [Padhan & Pandit (2023*b*)] that has shown how to use DNSs of the CHNS3 system (7.7)-(7.9) and their 2D counterparts to study the spatiotemporal evolution of the merger of liquid lenses in both 2D and 3D. In Fig. 20 we present illustrative results from these DNSs in 2D (top two rows) and 3D (bottom two rows). In 2D we give pseudocolor plots of ω, with overlaid velocity vectors in (a) the viscous regime and (b) the inertial regime; the $c_1 = 0.5$ contour (magenta line) indicates the lens interface; ω is normalized by its maximal absolute value for ease of visualization. Isosurface plots of c_1 (green) and $|\omega|$ (brown) for (c) the viscous regime and (d) the inertial regime. In the viscous regime, i.e., at large values of the Ohnesorge number Oh, a vortex quadrupole dominates the flow in the region of the neck in both